

Musa Azeem

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Education

University of South Carolina Honors College

Bachelor of Science in Computer Engineering; Overall GPA: 4.000

Summa Cum Laude; President's Honor List, all semesters

Columbia, SC

May 2024

University of South Carolina

Masters of Science in Computer Science; Overall GPA: 4.000

Columbia, SC

May 2025

Experience

Cognito Forms

Full-stack Software Engineer

Columbia, SC

Jun 2025 – Present

- Develop and maintain the Cognito Forms web app across a .NET Core backend and TypeScript/JavaScript + Vue.js frontend.
- Deliver bug fixes and performance improvements, optimizing algorithms to reduce latency and memory usage.
- Develop innovative features, including integrated LLM-based and payment services using Stripe, PayPal, and Square SDKs.
- Run delivery workflows with Azure DevOps pipelines, automated agent tasks, and production diagnostics in Application Insights.

University of South Carolina

Artificial Intelligence Research Assistant

Columbia, SC

Dec 2021 – May 2025

- Led the research and implementation of deep learning models using PyTorch (LSTM, CNN, Autoencoders), achieving high accuracy in human activity recognition, rodent sleep status prediction, and protein structure prediction, resulting in multiple publications.
- Engineered an ETL pipeline to process EEG data from SQL databases for analysis and machine learning; deployed on a server for real-time processing.
- Developed a Python library and ML models to evaluate protein structure predictions without experimental structures, producing an accuracy metric comparable to RMSD.
- Built and deployed full-stack solutions including an Android WearOS app for real-time accelerometer streaming into an edge ANN and a Flask web app for prediction workflows.

SIOS Technology Corporation

AI Solutions Architect & Software Engineering Intern

Columbia, SC

May 2023 – Feb 2024

- Led end-to-end development of a RAG-powered AI chatbot to deliver accurate, summarized answers based on information pulled from documentation, Salesforce tickets, and internal notes. Developed with a quantized local Llama-3 instance hosted via llama.cpp, an auto-updated vector database, and website/documentation parsing.
- Created a Python library to parse customer logs, extract critical information, detect error patterns, and present results through an interactive web interface.

Capgemini

Rise Intern - Technology Consultant Role

Houston, TX

Aug 2023

- Directed a rapid turnaround case study to deliver a mobile app launch strategy, functional prototype, and business model.
- Engaged with industry experts to gain insights into software engineering and global technology consulting processes.

Skills

Programming Languages: Python, C#, C/C++, TypeScript, JavaScript, R, Kotlin, Java, MATLAB, SQL, Bash, SystemVerilog

Tools/Libraries: PyTorch, TensorFlow, .NET, Pandas, NumPy, Flask, Matplotlib, Biopython, React, Vue.js, Express, Android SDK

Machine Learning: Reinforcement Learning, Decision Trees, K-Means Clustering, Model Evaluation & Diagnostics

Deep Learning: MLP, CNN, LSTM, Residual Networks, Autoencoder Architecture, Model Regularization

Data Analysis: Data Cleaning, Data Visualization, Exploratory Data Analysis (EDA), SQL Querying, ETL Pipelining

Other: Claude Code, Generative AI, Git, Linux, AWS, Docker, Embedded Systems, Microcontroller, Circuit Design, Chip Design

Publications

- [1] Musa Azeem and Homayoun Valafar, “Dihedral Angle Adherence: Evaluating Protein Structure Predictions in the Absence of Experimental Data,” in 2024 Congress in Computer Science, Computer Engineering, & Applied Computing (CSCE), Las Vegas, NV, USA, 2024. <https://doi.org/10.48550/arXiv.2407.18336>
- [2] Andrew Smith, Musa Azeem, et al., “Toward Concurrent Identification of Human Activities with a Single Unifying Neural Network Classification: First Step,” Sensors 24, no. 14: 4542. <https://doi.org/10.3390/s24144542>
- [3] Grant King, Musa Azeem, Savannah Noblitt, Ramtin Zand, and Homayoun Valafar, “Rep Smarter, Not Harder: AI Hypertrophy Coaching with Wearable Sensors and Edge Neural Networks.” <https://doi.org/10.48550/arXiv.2512.11854>

Awards & Engagement

NASA STEM Outreach Award Aug 2023 – May 2024
Secured funding to conduct research and lead eight outreach events, teaching high school students across South Carolina the basics of web programming. Developed a universal smartwatch application to record sensor data and facilitate event annotation.

Outstanding Senior in Computer Engineering Mar 2024
Recognized for major achievement in service, leadership, academics, and research.

Projects

Deep Learning Eating Detection (github.com/musa-azeem/eating-detection) Jul 2023 – May 2024
Researched ML techniques to use tri-axis accelerometer data to recognize eating gestures. Built PyTorch models including LSTM, CNN, Residual, and Autoencoder networks.

Protein Analysis (github.com/musa-azeem/protein-dihedral-visualization) May 2024 – May 2025
Developed a Python library and advanced data analysis and ML techniques to evaluate protein structure predictions in the absence of experimentally determined structures. Used to calculate an accuracy metric comparable to the industry standard RMSD, providing robust evaluation capabilities.

Faculty Dashboard Full Stack App (github.com/SCCapstone/K3MS) Aug 2023 – May 2024
Built a full-stack dashboard for UofSC faculty using Flask, React, and MySQL. Implemented REST APIs, CSV ingestion, data processing, and analysis; deployed with Apache.

Delta WearOS App (github.com/smithandrewk/delta) Jul 2022 – May 2024
Developed a Kotlin WearOS app to record, annotate, and process accelerometer data. Live-fed data into an edge neural network to detect smoking and supported data labeling via Jetpack Compose UI.